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Bioinspired, Biomimetic and Nanobiomaterials

Exploring metamaterials for healthcare: an extensive insight into capabilities and applications

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The fast growth of healthcare technologies has created a strong need for advanced materials that can support accurate diagnosis, less invasive treatments, and patient-specific solutions. Traditional materials often fail to meet these requirements because they are not easily adjustable and cannot adapt to changing biological conditions. In this context, metamaterials have emerged as a promising solution. These are specially designed materials whose properties depend on their structure rather than their chemical composition. Metamaterials can control electromagnetic, mechanical, acoustic, and thermal waves in unique ways, opening new possibilities in medical science. This paper presents a clear and practical overview of the use of metamaterials in healthcare. It explains their basic concepts, design methods, and importance in modern medicine. The study highlights their applications in diagnostics, treatment, and biomedical devices, where they offer high sensitivity, targeted energy delivery, better compatibility with the human body, and real-time monitoring. The paper also discusses the integration of metamaterials with artificial intelligence, enabling smart and personalized healthcare solutions. While metamaterials show great potential in improving disease detection and treatment outcomes, challenges such as complex fabrication, scalability, biocompatibility, and regulatory approval still need attention. This work provides useful insights for researchers and healthcare professionals.

Keywords: additive manufacturing/artificial intelligence/biomaterials/biomedicine/healthcare innovation/medical diagnostics/metamaterials/nanomaterials/precision medicine/smart biomaterials/UN SDG 3: Good health and well-being/UN SDG 9: Industry, innovation and infrastructure/UN SDG 12: Responsible consumption and production

1. Introduction

The global healthcare system is undergoing a paradigm shift due to demographic changes, the growing disease burden, and increasing demand for customized and preventive care. Diagnostic and therapeutic technologies available today have never been burdened more as the entire world has been stricken with a range of chronic diseases that include cancer, cardiovascular disorders, musculoskeletal degeneration, and neurological disorders. Conventional healthcare materials and devices, although effective to some degree, may not be flexible, sensitive, and multifunctional enough to meet current clinical demands.[1]

Early and precise diagnosis is one of the greatest problems of modern medicine. Traditional imaging and sensing methods often have poor resolution, low sensitivity to disease markers at early stages, and limited specificity. In the same way, most treatment modalities are based on invasive methods or are systemic in drug delivery, resulting in unwanted side effects and poor patient compliance. Mechanical imbalance between implants and biological tissues is usually a cause of stress shielding, implant loosening and failure in the long term in orthopedic and implantable devices.[2–4]

Metamaterials have become a disruptive category of engineered materials in this landscape, as they can overcome several inherent shortcomings of more traditional materials. Metamaterials are man-made structured materials, the extraordinary properties of which are due not only to their chemical composition but also to their designed architecture. Metamaterials can produce properties unavailable in nature, including negative refractive index, negative permittivity, negative permeability, auxetic mechanical behavior, and highly localized energy concentration by adjusting the geometry, size, and arrangement of subwavelength unit cells.[5–7]

Metamaterials have also achieved widespread use since they were theoretically predicted by Veselago in the late 1960s and experimentally demonstrated decades later in various fields, including electromagnetics, acoustics, mechanics, and thermal management.[8–10] Later,

their use in the medical field has been growing with significant traction, as there is a dire necessity to introduce high-performance materials that can interact with biological systems in an intelligent manner.[11]

Medical imaging utilizes metamaterials, which enable better control of wave propagation, resulting in high spatial resolution, a high signal-to-noise ratio, and compact imaging devices. Metamaterial-based systems, biosensors, and hyperthermia systems offer noninvasive, localized, and highly sensitive alternatives to traditional methods used in cancer diagnosis and treatment. Mechanical metamaterials have been used in orthopedics to design implants that closely resemble the anisotropic and hierarchical bone structure to minimize stress shielding and enhance the process of osseointegration.[12–15]

Metamaterials further enhance their applicability in healthcare through integration with emerging technologies, including additive manufacturing, nanotechnology, and artificial intelligence (AI). Additive manufacturing enables the design of patient-specific metamaterials, and AI techniques and optimization, as well as soft computing, are utilized to enhance material performance, predict biological interactions, and inform clinical decision making. This merging establishes metamaterials as a foundation for third-generation smart healthcare systems.[16–19] The clinical translation of the metamaterials is not very wide, even though it has huge potential. There are still issues perceived as complex, such as scalability in fabrication, long-term biocompatibility, reproducibility, and regulatory approval, which impede the large-scale use of these technologies. In addition, interdisciplinary disconnects exist between materials scientists, clinicians, and regulatory agencies, which tend to hinder the translation of laboratory prototypes into real-world medical devices.[20–23]

This paper aims to critically and systematically review metamaterials in healthcare, with a particular emphasis on clinical responsiveness and translational capability. The paper presents a critical analysis of the main principles of metamaterials, the reasons why they should be utilized in the healthcare sector, and their role in addressing the most pressing medical challenges. The following sections discuss key metamaterials, smart metamaterials, classical applications, and a broad range of applications, and then critically examine the shortcomings of metamaterials, outlining future research directions.

2. Metamaterials

Metamaterials are artificially fabricated materials, which are built in unit cells (periodically or a periodically arranged) of a size smaller than the wavelength of the external stimulus on which the material responds. In contrast to more traditional materials, where the primary contribution to properties is made by the chemical composition, metamaterials derive their effective physical properties from their geometry and spatial arrangement.[24–26] 1D, 2D, and 3D metamaterial have been shown in Figure 1.

The hallmark of metamaterials is their ability to control various types of energy, including electromagnetic, mechanical, acoustic, and thermal energy, in unconventional ways. Incidentally, electromagnetic metamaterials may curve light in reverse, focus electromagnetic energy above the diffraction limit or absorb certain frequencies preferentially. Negative Poisson's ratio, ultra-low stiffness with high strength and programmable deformation behavior can be achieved with mechanical metamaterials.[27–30]

These properties can be applied in a healthcare context because the heterogeneous and sensitive issue of biological systems to external stimuli is inherent in them. Classical materials are usually imprecise enough that they do not interact sufficiently with tissues, cells, and biomolecules. The metamaterials themselves, however, are scale- and frequency-insensitive and can be adjusted to couple with biological processes, enabling more efficient interaction and control.[31–34]

2.1 Evolution and classification of metamaterials

Metamaterials are developmental and categorized based on their properties, which are developed to show attributes that are not possible using natural materials that can possess the identical atomic arrangement as them. Metamaterial development can be divided into three distinct periods. The early days were focused on developing an electromagnetic metamaterial for use in both microwave and optical applications. The second step expanded to acoustical and mechanical metamaterials, where sound and mechanical stress could be controlled. The multifunctional and intelligent metamaterials, featuring frequent sensor correspondence, actuators, and intelligent algorithms, are the primary features of the current stage.[35–40] Figure 2 represents the classification of metamaterials for structural study. Metamaterials that apply to healthcare can be categorized depending on their predominant physical response:

- Electromagnetic metamaterials: imaging, biosensing, antenna, and radiation shielding.
- Mechanical metamaterials: orthopedic implant and prosthetic use.
- Acoustic metamaterials: experts in ultrasound imaging and therapeutic acoustics.
- Optical and plasmonic metamaterials: biomolecular sensing and bioimaging.
- Terahertz metamaterials: noninvasive diagnostics and drug monitoring.

This categorization emphasizes the fact that metamaterials are multidisciplinary and can be used in healthcare in a wide range.[41]

2.2 Need for metamaterials in healthcare

The use of metamaterials in medical practice is driven by several unmet clinical and technological requirements that cannot be sufficiently met by traditional materials.

2.2.1 Limitations of conventional medical materials

Conventional medical materials, including metals, polymers, and ceramics, are commonly specialized for single-use applications. For

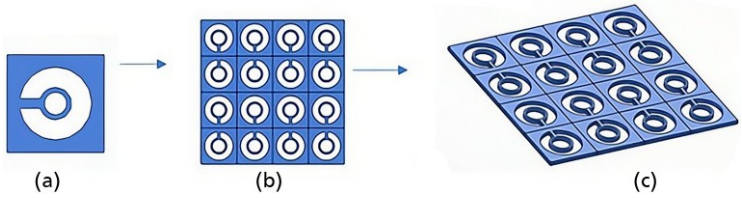


Figure 1. Representation of a square split-ring resonator: (a) 1D metamaterial, (b) 2D metamaterial, and (c) 3D metamaterial

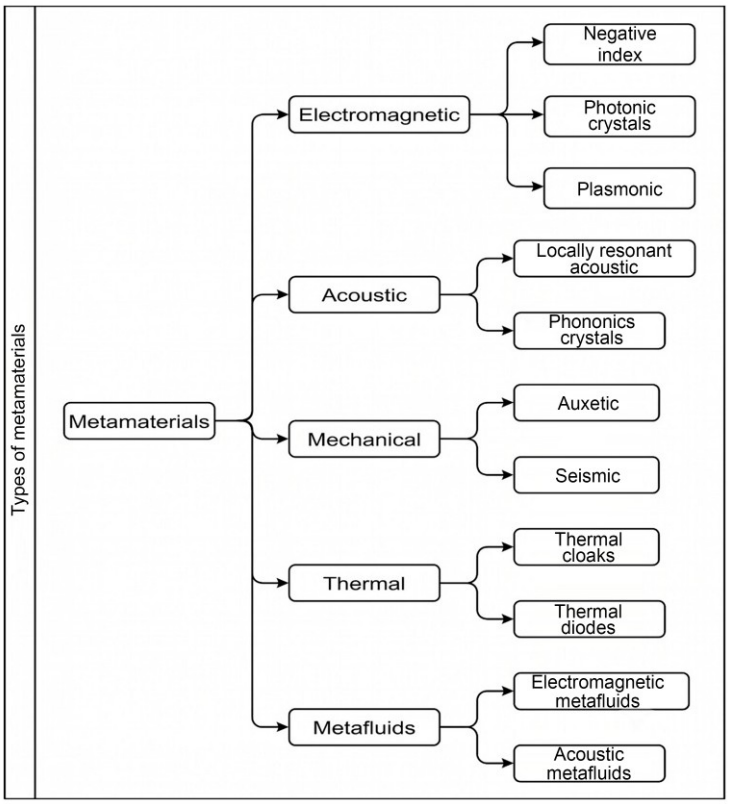


Figure 2. Classification of metamaterials

example, metallic implants are stronger but less biomechanically compatible, whereas polymeric materials are more flexible but have lower durability. On the same note, traditional imaging materials are not tunable and cannot easily be made to produce high-quality images without expanding the size or power consumption of the system.[42–43] Such constraints lead to poor quality of diagnostics, higher chances of implant failure, and patient discomfort.

2.2.2 Demand for precision and personalization

Precision medicine has become one of the main concepts in modern healthcare, with the diagnosis and treatment process being customized to the specific patient features. Such personalization is made possible by metamaterials, which can be used as the guiding design parameter of control over material properties, not composition. Advanced manufacturing methods can be utilized to produce patient-specific metamaterial implants, sensors, and therapeutic devices, leading to enhanced clinical outcomes.[44–45]

2.2.3 Noninvasive and targeted therapies

Some examples of noninvasive treatment modalities that utilize metamaterials include microwave hyperthermia, photothermal therapy, and targeted drug delivery. It is necessary because metamaterials can concentrate energy or therapeutic factors at the site, causing the least possible damage to normal tissues while also minimizing side effects. This is particularly important during the treatment of cancers and chronic diseases.[46–47]

2.2.4 Real-time monitoring and smart healthcare

Metamaterials can be utilized to enhance wireless sensing and smart systems, enabling the continuous monitoring of physiological measurements and determining the integrity of the implant and the recovery process. These skills are required for preventive care, early intervention, and long-term disease management.[48] In conclusion, the need to develop metamaterials in healthcare is driven by the increasing demand for materials that are adaptive, possess multifunctional properties, are patient-specific, and can be easily integrated with smart healthcare systems. They possess distinctive features that make them essential enablers of next-generation medical technologies.[49]

2.3 Major metamaterials used in healthcare

The current blistering growth of metamaterial studies in healthcare has made numerous classes of materials optimized toward clinical applications. Metamaterials, in contrast to traditional biomaterials, are no longer determined by composition alone, but by the engineered architecture, and tuning in electromagnetic, mechanical, acoustic, and optical regimes, where none have ever been tuned to as large an extent. This section will review the key metamaterial types currently used in healthcare or proposed for use in healthcare, with a focus on their mechanisms and clinical applications.[50–51] Table 1 represents the major metamaterials used in healthcare.

2.3.1 Electromagnetic metamaterials

One of the widely studied classes of electromagnetic metamaterials in healthcare is the electromagnetic class. They are designed with controlled electromagnetic wave propagation by special permittivity and permeability (which is usually negative or near-zero over a range of frequencies). The electromagnetic metamaterials described by such properties can bend, focus, absorb, or filter electromagnetic waves with high accuracy.[52] Electromagnetic metamaterials are utilized to enhance the performance of magnetic resonance imaging (MRI), microwave imaging, and terahertz imaging in the field of medical imaging. These materials enhance the signal-to-noise ratio (SNR) and spatial resolution by concentrating electromagnetic fields and improving impedance matching, while reducing power input and system size to significantly improve spatial resolution.[53] Metamaterial-based resonators are utilized in biosensing to detect small variations in dielectric properties resulting from biomolecular interactions, enabling the detection of disease biomarkers at ultrasensitive limits.[54–55]

2.3.2 Mechanical metamaterials

Mechanical metamaterials exhibit anomalous mechanical behaviors that depend on the geometry of the material, rather than the material itself. The possibility to obtain a negative Poisson's ratio (auxetic behavior), that is, the material gets widened on the side when stretched and narrow on the side when compressed, is one of the most clinically important properties of mechanical metamaterials.[56] Mechanical metamaterials can be used in orthopedics to create implants that are anisotropic and hierarchical in structure, like natural bone. This reduces mechanical incompatibility, stress shielding, and improves osseointegration, thereby increasing the lifespan of the implant and enhancing patient outcomes.[57–60] Patient-specific mechanical metamaterials, tailored to meet specific anatomical and biomechanical needs, have also become possible with the advent of additive manufacturing.

2.3.3 Acoustic and phononic metamaterials

Phononic or acoustic metamaterials are artificial materials designed to control sound and elastic waves. Such materials can be characterized as having negative mass density or negative bulk modulus, allowing for a variety of other unique phenomena, including acoustic band gaps and wave focusing.[61] The application of acoustic metamaterials in healthcare is a growing field of research, particularly in ultrasound imaging and therapy. These materials enhance image resolution and provide treatment with acoustic energy, which can be directed by improving wave confinement and directionality, thereby allowing focused ultrasound ablation to be performed.[62] Their ability to filter undesirable vibrations also makes them suitable for reducing noise in medical equipment.

2.3.4 Optical and plasmonic metamaterials

Optical metamaterials operate at visible and near-infrared frequencies, and at shorter wavelengths, most of them utilize plasmonic effects. The materials can support the process of superlensing, higher absorption, and surface-enhanced Raman scattering.[63] Optical metamaterials are also utilized in biomedical diagnostics, including label-free biosensing, molecular detection, and high-resolution bioimaging. They possess a high ability to amplify low-intensity optical signals, making them quite helpful in the detection of early-stage diseases, including cancer diagnostics.[64]

Table 1. Types of metamaterials used in healthcare

Type of metamaterial	Description
Electromagnetic metamaterials	Control electromagnetic wave propagation for imaging, sensing, antennas, and radiation shielding
Mechanical metamaterials	Exhibit programmable stiffness, auxetic behavior, and bone-mimicking mechanical properties
Acoustic metamaterials	Manipulate sound and elastic waves for enhanced ultrasound imaging and therapy
Optical metamaterials	Enable subwavelength light control for biosensing, bioimaging, and diagnostics
Terahertz metamaterials	Operate in the THz range for noninvasive diagnostics and drug monitoring
Nanometamaterials	Combine nanoscale features with metamaterial concepts for targeted therapy and sensing

2.3.5 Terahertz metamaterials

Metamaterials in the frequency range of microwaves and infrared radiation are referred to as terahertz (THz). Among the reasons that make this spectral range of interest in biomedicine, THz waves are nonionizing and can detect the water content and molecular composition.[65] THz metamaterials have been used in skin cancer detection, drug release, and biological tissue characterization. That can be explained by the high sensitivity of this method to minute biochemical changes and the fact that it can be used in noninvasive and real-time diagnostics.[66]

2.4 Smart metamaterials for healthcare

Metamaterials turning into smart and intelligent systems is a crucial innovation in the field of healthcare. Intelligent metamaterials can detect, respond to, and adaptively respond to external stimuli in the form of mechanical stress, temperature changes, electromagnetic flux, or biochemical cues.[67]

2.4.1 Integration with sensors and actuators

Embedded sensors and actuators can be added to intelligent metamaterials, enabling them to respond to feedback and act adaptively. As an example, orthopedic metamaterial implants with wireless strain sensors can also be used to measure the mechanical loads and the healing process without being invasive.[68]

2.4.2 Role of artificial intelligence and soft computing

The trend of AI and soft computing techniques is crucial in enhancing the functionality of smart metamaterials. The optimization of unit cell design, prediction of material behavior under physiological conditions, and individual optimization of device performance to specific patients is performed with the help of machine learning algorithms.[69] AI-powered metamaterial systems facilitate intelligent decision making in clinical contexts by comparing sensor data with patient-specific health indicators. The integration is especially useful in cancer therapy planning, implant monitoring and adaptive imaging systems.[70]

2.4.3 Adaptive therapeutic systems

The dynamically controlled parameters of treatment enable the use of smart metamaterials and adaptive therapy, in which the parameters of the therapeutic element are adjusted in response to real-time feedback. In microwave hyperthermia, for example smart metamaterial lenses can be used to control the distribution of electromagnetic fields to keep the tumor tissue as hot as possible and to avoid overheating healthy tissues.[71–72]

2.5 Classical capabilities of metamaterials in the healthcare sphere

Although the industry is constantly advancing in its innovativeness, several traditional capabilities of metamaterials form the basis of healthcare applications. These functions are based on fundamental physical processes and remain clinical motivators.

2.5.1 Electromagnetic wave manipulation

Probably the most important characteristic of metamaterials is the ability to control the electromagnetic waves on a very fine level. It possesses the ability to absorb, cloak, and focus waves. The above capabilities in the healthcare sector have been transformed to deliver better quality imaging, improved diagnosis, and focused energy delivery.[73–74]

2.5.2 Mechanical compatibility and load management

Mechanical metamaterials allow the behavior of stiffness, strength, and deformation to be controlled precisely. This enables the implants to distribute mechanical loads in a more even manner, where stress concentrations are minimized, and natural bone remodeling is encouraged.[75]

2.5.3 Miniaturization of medical devices

Metamaterials enable the reduction in size of medical devices, including antennas, sensors, and imaging units. Small-sized devices are especially beneficial in minimally invasive surgery, implantable surgery, and wearable health technologies.[76]

2.5.4 Energy localization and selective absorption

Metamaterials can be used to perform localized therapy through selective absorption or concentration of energy in designated areas. This ability will be crucial in the application of hyperthermia cancer therapy and photothermal therapy, where the efficacy of the treatment is determined by the precision of energy delivery.[77–79]

2.5.5 High sensitivity and selectivity

Sensors made of metamaterials are particularly sensitive to environmental changes, such as changes in dielectric properties, strain, or biochemical composition. This sensitivity allows detection of disease at an early stage, as well as in real-time physiological processes.[80]

2.6 Broad applications of metamaterials in healthcare

Metamaterials can be utilized in healthcare in various ways, and their practical implications can be illustrated through the diverse applications of metamaterials. The solutions to the major issues in diagnosis, therapy, rehabilitation, and patient monitoring can be achieved with the help of metamaterials, in which electromagnetic, mechanical, acoustic, and thermal interactions can be controlled accurately. The section also provides an overview of the diverse applications of metamaterials in healthcare, including a discussion of the key areas.[81–85] Broad application areas of metamaterials in healthcare sector is shown in Table 2 and major contribution of metamaterials in different medical sectors is shown in Table 3 in brief.

Table 2. Broad application areas of metamaterials in the healthcare sector

Application area	Description
Medical imaging	Enhanced resolution and sensitivity in MRI, microwave, and terahertz imaging
Cancer diagnosis	Ultra-sensitive biosensors for early cancer detection
Cancer therapy	Targeted hyperthermia, photothermal, and microwave-based treatments
Orthopedic implants	Bone-mimicking implants with reduced stress shielding
Radiation shielding	Lightweight protection for patients and medical personnel
Wireless health monitoring	Implantable and wearable metamaterial-based sensors
Drug delivery	Targeted and controlled therapeutic release systems
Rehabilitation devices	Adaptive mechanical support and motion control

Table 3. Contribution of metamaterials in healthcare

S.N.	Implant / device / material	Healthcare field	Quantitative metric (reported)	Contribution	Ref.
1	Wireless strain sensor with SRRs	Wireless strain measurement	Sensitivity 1.09 kHz/kgf (nested design)	Low nonlinearity errors; first nested-metamaterial strain sensor	24
2	Terahertz metamaterial-based sensor	Wireless strain measurement	Sensitivity 320 GHz/μm	Detection of bi-axial strain; ~3.24 × over prior THz sensor	78
3	Intramedullary nails	Orthopedic	Gap 0.15–0.60 mm; 5%–20% interfragmentary strain	Mitigates metal–bone stiffness mismatch via 3D-printed cylindrical metamaterials	9
4	Auxetic meta-biomaterial implant (Ti)	Orthopedic	Elastic modulus 84.2 MPa–11.1 GPa; yield strength 6.9–279.9 Mpa	Tunable stiffness matched to bone range; >300 specimens evaluated	10
5	Graphene and its derivatives	Cancer treatment	—	Better specificity and fewer side effects in treatment	25
6	Metamaterial biosensor (PMMA substrate)	Cancer detection	Sensitivity 658 nm/RIU (TiO_2); FOM 2431	Label-free early cancer-cell detection	27
7	Microstrip sensor integrated with AMC	Cancer detection	Gain 8.5 dBi at 4.56 GHz	Early-stage breast, lung, and brain cancer detection	28
8	Petahertz dual-band absorber	Skin cancer detection	Sensitivity up to 440.5 THz/RIU; bands at 0.909 and 1.215 PHz	Early detection of nonmelanoma skin cancer	29
9	Multi-spectral focal plane arrays	Medical imaging	—	Upgraded THz imaging enabling real-time imaging	42
10	Metamaterial absorber RF shield	Radiation shielding / MRI	Transmit efficiency + 32%; peak SAR −22%	Miniaturized RF shield operating at 7 T/9.4 T	62
11	Gold nanoparticles (AuNPs)	Cancer detection	—	Surface enhancement in Raman spectroscopy and photothermal therapy (PTT)	35
12	Acoustic metamaterial transdermal patch	Drug delivery	~9.3 × more transdermal delivery vs. passive; 2.5 g/burst epinephrine; resonance ~1.06 MHz	On-demand, dose-controlled transdermal delivery; reversed anaphylaxis in vivo	88
13	Silk-fibroin THz metamaterial patch	Drug delivery / monitoring	Resonance ~1.3 THz	Biodegradable patch; real-time resonance-based drug-release monitoring	86
14	3D-printed calcium phosphate metamaterial	Additive manufacturing / bone	Sub-300 nm print resolution; 5 nm prenucleation clusters	Nanoscale 3D printing of bioinspired CaP for bone and drug-delivery scaffolds	89

2.6.1 Metamaterials in orthopedic and biomedical implants

Orthopedic implants are among the healthcare applications of metamaterials that have the greatest impact. Traditional metallic implants, such as titanium alloys and stainless steel, have elastic moduli that are significantly greater than those of natural bone. Such a mechanical mismatch causes stress shielding, which decreases the transfer of loads to the surrounding bone tissue and

ultimately leads to bone resorption and implant loosening.[86] The mechanical metamaterials provide a radical solution through which mechanical stiffness and anisotropic behavior can be programmed. Implants can be made to mimic hierarchical and directionally dependent mechanical properties of bone through architected lattice structures.[87] The stability of implants in specific cases is enhanced by auxetic metamaterials, which provide greater contact with the surrounding tissue, leading to the even distribution of stress.

- Clinical advantages
- Improved osseointegration
- Reduced implant failure rates
- Enhanced long-term biomechanical compatibility
- Patient-specific implant customization via additive manufacturing

According to recent clinical research, implants based on metamaterials could significantly enhance the implant life cycle and reduce the need for revision surgery.[88]

2.6.2 Cancer diagnosis using metamaterials

Cancer is a serious disease that requires early detection to increase the chances of survival. Nonetheless, the traditional diagnostic instruments are not usually sensitive enough to identify malignancies at an early stage. The solution to this problem, as proposed by metamaterial-based biosensors, is to utilize resonant electromagnetic and optical responses to detect the smallest modifications in biological samples.[89]

Metamaterial biosensors: Metamaterials, such as electromagnetic and plasmonic metamaterials, are constructed to exhibit sharp resonance peaks that change in response to variations in dielectric properties during biomolecular binding events. This enables the detection of cancer biomarkers at an extremely low concentration, promoting early diagnosis.[90]

Noninvasive screening: Terahertz metamaterials have nonionizing and noninvasive diagnostic applications, making them suitable for skin and breast cancer screening. Their water sensitivity and tissue composition provide a valuable contrast for imaging between healthy and diseased tissues.[91] Figure 3 presents the biosensing mechanism based on metamaterial resonators, showing the resonance frequency shift induced by biomolecular binding.

2.6.3 Metamaterials in cancer therapy

In addition to diagnosis, metamaterials play a crucial role in cancer treatment, particularly in hyperthermia and photothermal therapies. These treatments are based on the controlled heating of tumor tissues, which causes cell death while sparing the surrounding healthy tissues.[92]

Microwave and electromagnetic hyperthermia: Electromagnetic energy can be focused in tumor areas using metamaterial lenses and absorbers and achieve uniform and controlled heating. This

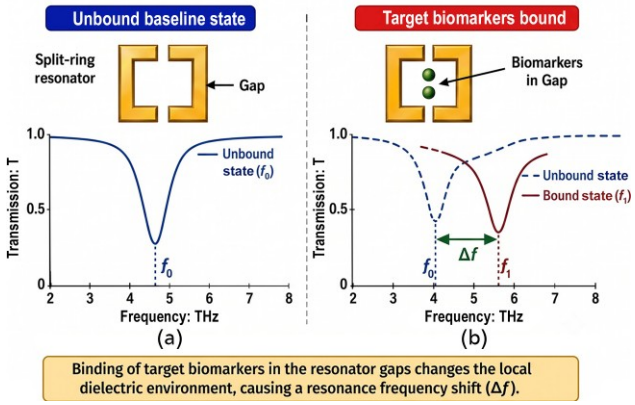


Figure 3. Representation of metamaterial biosensor: (a) unbound baseline state and (b) target biomarkers bound

precision is superior to traditional hyperthermia methods in terms of treatment and reduced negative outcomes.[93] Figure 4 demonstrates the principle of metasurface-assisted electromagnetic hyperthermia, illustrating the focusing of electromagnetic energy for localized tumor heating.

Photothermal therapy: The most effective nanoconversion of light into heat can be achieved using nano metamaterials, in particular plasmonic nanostructures. With targeting agents functionalized on them, these materials allow the selective destruction of cancer cells.[94,98]

2.6.4 Medical imaging applications

Modern healthcare is inseparable from medical imaging, and the use of metamaterials is a crucial means of enhancing the performance of imaging in various modalities.

Magnetic resonance imaging: Metamaterials can be used to improve the performance of MRI by improving the homogeneity of radiofrequency fields and SNR. This results in an increased level of image resolution, as well as a reduction in scanning time, which enhances patient comfort and diagnostic accuracy.[98–101]

Microwave and terahertz imaging: Antennas utilizing metamaterials enable compact, high-resolution microwave and terahertz imaging. Lenses based on metamaterials enable compact and high-resolution imaging in the microwave and terahertz frequency ranges. The systems have proven to be especially useful in the detection of breast cancer and characterization of tissues.[42,43,102] Figure 5 illustrates the working principle of a metamaterial superlens for subwavelength imaging, highlighting the amplification of evanescent waves and image reconstruction beyond the diffraction limit.

2.6.5 Radiation shielding in healthcare

Radiation is very dangerous to patients and medical workers. Traditional shielding materials, such as lead, are dense, poisonous, and hard.

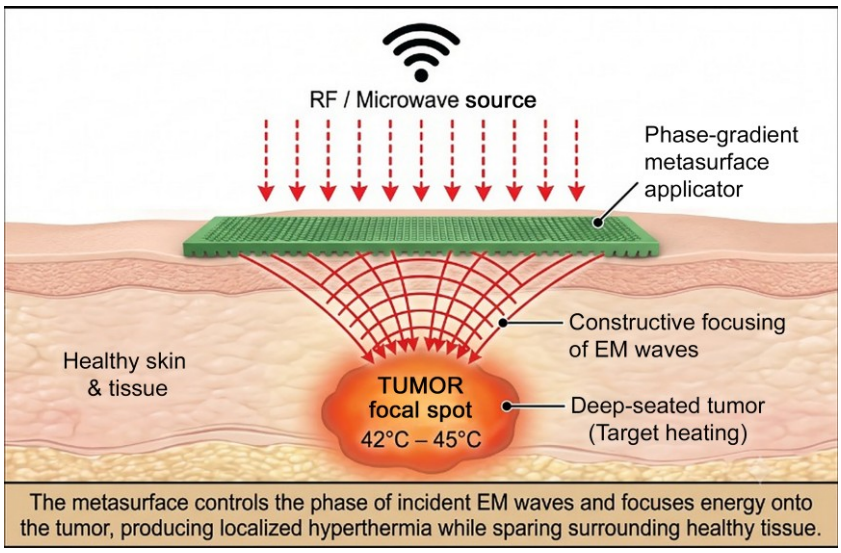


Figure 4. Metamaterial assisted-electromagnetic hyperthermia

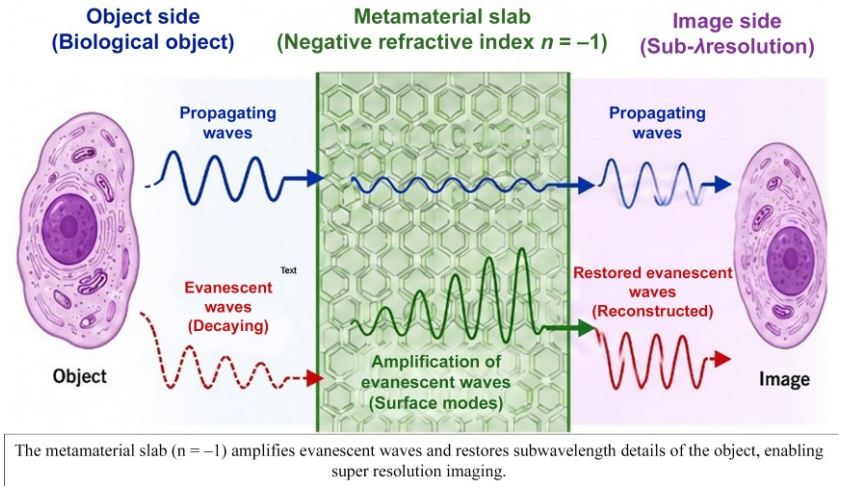


Figure 5. Metamaterial superlens for subwavelength imaging

Metamaterial-based shielding: Lightweight and flexible radiation shielding is provided by electromagnetic metamaterials. These materials provide proper protection by absorbing or deflecting specific frequencies of radiation, thereby enhancing comfort and mobility.[61,63]

2.6.6 Wireless health monitoring and wearable devices

Telemedicine and remote healthcare have created a need for wearable and implantable healthcare monitoring devices. Metamaterials are used to make compact, energy-efficient, and sensitive wireless sensors.[79]

Implantable sensors: Strain sensors and pressure sensors are implantable sensors based on metamaterials that are used to monitor mechanical loads, progress of healing, and implant integrity in real-time.[24]

Wearable healthcare systems: Antennas made of metamaterials enhance the efficiency of wireless communication in wearable devices, enabling continuous monitoring of patient health without compromising their comfort.[61]

2.6.7 Targeted drug delivery systems

The delivery of drugs to the targeted site reduces the side effects associated with systemic administration of the drug and achieves maximum therapeutic effects. Stimuli-responsive and controlled

drug delivery is made feasible using metamaterials at the nanoscale.

Stimuli-responsive metamaterials: Nano metamaterials are intelligent and react to external stimuli, such as temperature, pH, or electromagnetic fields, and deactivate drugs at the target site. The method is particularly noteworthy in the treatment of cancer and chronic diseases.[38,39,88] Figure 6 depicts the concept of stimulus-responsive drug delivery, where metamaterial structures regulate controlled drug release under external excitation.

2.6.8 Rehabilitation and assistive devices

The adaptive mechanical support and motion control is another field in which metamaterials are applied to rehabilitation engineering.

Adaptive prosthetics and orthoses: Mechanical metamaterials have also enabled the development of lightweight and energy-efficient prosthetic devices with adjustable stiffness and damping properties, allowing patients undergoing rehabilitation to achieve a higher level of mobility and comfort.[4,13]

3. Discussion

The rapid development of metamaterials has become a disruptive technology in the healthcare sector. Metamaterials have been applied in diagnostics, therapeutics, implants, and monitoring systems, offering an unprecedented level of control over physical interactions at scales crucial to biological systems. The case in this segment involves a critical analysis of the translational significance of metamaterials, their clinical readiness, and the challenges that need to be overcome to achieve the full potential of metamaterials in healthcare. The design-directed functionality of metamaterials is one of the most significant benefits. Unlike conventional biomaterials, metamaterials can be tailored to meet the required clinical requirements by merely adjusting their structural geometry, but not their chemistry. It is a paradigm shift that is directly related to the concepts of precision medicine, where a customized solution is offered to the patient in terms of imaging, therapy, and implants. Mechanical metamaterials. The bone-mimicking stiffness gradients have been shown to enhance the osseointegration and minimize stress shielding effects, which overcomes a long-standing orthopedic surgery challenge.

Metamaterial-based biosensors and imaging devices have consistently demonstrated higher sensitivity and specificity in diagnostic applications compared to traditional technologies. The ability of electromagnetic and optical metamaterials to amplify weak signals enables the detection of diseases, such as cancer, at an early stage, where early detection can significantly improve prognosis. Likewise, terahertz metamaterials can also be used to offer nonionizing, label-free diagnostic functionality, which is especially applicable to soft tissue analysis.

It is essential to note that, although these benefits exist, metamaterials in their clinical translation are still in their infancy. The literature primarily presents research types of in vitro studies, animal studies, or provisional evidence. Huge clinical trials and long-term performance assessments are not common. The existence of this gap emphasizes enhanced cooperation among the material scientists, biomedical engineers, clinicians, and regulatory authorities.

Another crucial aspect is the incorporation of metamaterials into the current healthcare system. Despite the devices that are founded on the utilization of the metamaterials offering better performance, their application will be dependent on their compatibility with the existing imaging practice, surgery, and regulatory practices. The conversion of laboratory prototypes into clinically approved devices also requires intensive validation, repeatability, and adherence to high medical standards.

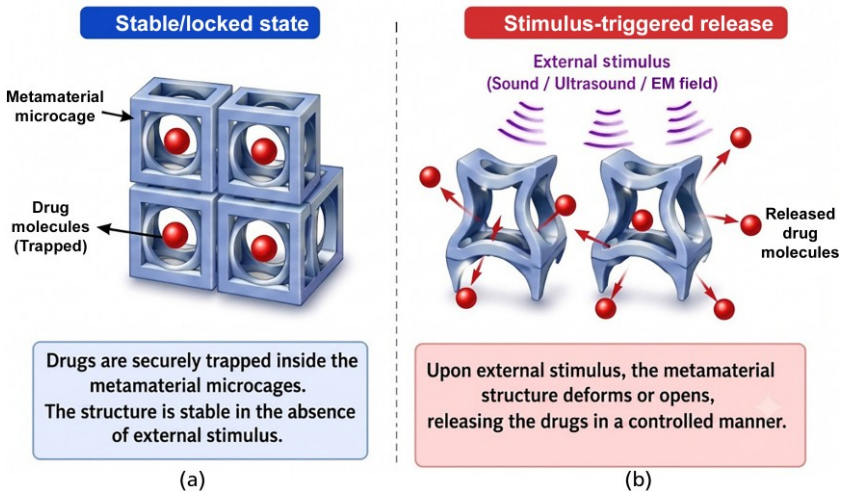


Figure 6. Metamaterial drug delivery: (a) stable/locked state and (b) stimulus-triggered release

4. Limitations of metamaterials for healthcare

The potential of metamaterials is excellent; however, several limitations render them inaccessible to the entire healthcare sector. These problems are all technical, biological, economic, and regulatory, and should be tackled in a stepwise manner to achieve successful clinical translation. Table 4 represents major limitations of metamaterials in field of healthcare.

4.1 Fabrication and scalability challenges

The fabrication of metamaterials is typically based on the use of high-tech production techniques, such as lithography, laser sintering, and high-resolution additive manufacturing. Although these methods enable refined control of the geometry, the production cost also increases, and the process can be scaled. Another significant challenge that has not been achieved at a clinical level is consistency of quality.

4.2 Biocompatibility and safety concerns

The biocompatibility requirement for implantable and wearable healthcare applications is urgent. In most metamaterials, nanostructures, metals, or polymers, their long-term interaction with biological tissues has not been adequately investigated. Among the possible issues are inflammation, immune response, and material degradation. These issues are to be resolved through the conduct of extensive biocompatibility testing and long-term clinical research.

4.3 Regulatory and standardization issues

There are complex regulations for medical equipment that uses metamaterials. Approvals cannot be easily granted as the metamaterial-based devices do not have standardized testing. Regulatory bodies are strict in requiring a broad scope of evidence on safety, efficacy, and reliability, which can be very slow in terms of commercialization.

5. Future scope of metamaterials in healthcare

The future of materials in healthcare is promising, despite the challenges it presents. Next-generation limitations are likely to be overcome by new research and the convergence of technology, leading to new opportunities.

5.1 Ai-assisted metamaterial design

AI and ML will become key to the future of metamaterials. The integration of rational microarchitectural design principles together with machine learning and AI tools holds strong promise for developing multifunctional meta-implants with unique properties.[15] Design optimization based on AI can accelerate the discovery of new metamaterial architectures for medical applications, and the employment of AI in the design process can elevate personalization further – enabling implants that not only fit a patient precisely but also actively respond to dynamic changes in the tissue environment.[15] In hospitals, AI-driven metamaterial systems can assist in adaptive diagnostics and customized therapy planning.[69,100] However, the convergence of AI and intricate materials science also raises ethical and regulatory considerations around long-term safety, data privacy, and machine-learning bias that must be addressed before clinical deployment.[15]

5.2 Additive and 4D printing for patient-specific design

Advances in additive manufacturing are enabling the creation of highly precise, complex metamaterial structures. Using specific anatomical data, patient-specific implants, prosthetics, and therapeutic devices can be produced to enhance clinical outcomes.[70] An emerging frontier is 4D printing, in which shape-morphing meta-implants are fabricated with programmable, time-dependent deformation; this requires novel designs and models to predict the bending behavior and stiffness of elements, and although creating multiscale meta-biomaterials remains challenging, their programmable shape deformation holds significant promise.[15,21] Material selection, surface biofunctionalization, drug-delivery mechanisms, and emerging 4D-printing techniques together represent critical areas for further advancement of meta-implant design.[15]

Additive manufacturing has become central to realizing the complex, subwavelength architectures that define metamaterials, enabling patient-specific implants and devices that conventional fabrication cannot achieve.[16–20] Layer-by-layer metal and polymer printing allows precise control over unit-cell geometry, porosity, and gradient structures, which is particularly valuable for orthopedic meta-implants that must mechanically match anisotropic bone tissue.[9,13,15] Beyond single-material printing, composite and multimaterial strategies further expand the design space: by combining constituents with differing stiffness, conductivity, or

Table 4. Major limitations of metamaterials for healthcare

Limitation	Description
Fabrication complexity	Advanced micro- and nanofabrication techniques increase cost and limit scalability
Biocompatibility	Long-term biological interaction and toxicity require extensive validation
Mechanical durability	Fatigue and long-term stability under physiological conditions remain uncertain
Scalability	Difficulty in mass production for clinical deployment
Cost	High manufacturing and material costs limit affordability
Regulatory barriers	Lack of standardized testing and approval frameworks
Integration challenges	Compatibility issues with existing medical devices and systems

degradation behavior, composite metamaterials can independently tune mechanical and functional properties to suit specific clinical requirements.[4,8,19] The integration of additive manufacturing with such composite and rationally designed biomaterials therefore offers a practical route toward scalable, multifunctional, and clinically translatable metamaterial devices.[17,22]

5.3 Bioresorbable and bioinspired metamaterials

Future research directions also include biodegradable and bioinspired metamaterials, which carry fewer long-term biological effects and are especially appealing for temporary implants, drug-delivery systems, and tissue engineering.[71] Extrusion-based printing of biodegradable metals such as magnesium, zinc, and iron-based alloys has been demonstrated, and biodegradable meta-implants have shown promising fatigue resistance in simulated body fluid.[15] Multimaterial additive manufacturing further enables biodegradable iron–manganese scaffolds that are MRI-compatible and biodegrade within the ideal range for bone substitution, while biofunctionalized porous scaffolds embedded with silver or iron nanoparticles can help prevent implant-associated infection and promote bone regeneration.[15]

5.4 Digital twins and smart, connected healthcare systems

Smart and connected healthcare systems will be achieved through the convergence of the Internet of Medical Things with metamaterials. Metamaterial-based sensors and antennas will support real-time monitoring, remote diagnosis, and predictive healthcare, aligning with the goals of preventive and personalized medicine.[72,79] Looking further ahead, multi-physics simulation frameworks that account for the spatial distribution of mechanical properties, cyclic-loading effects on biodegradation, and crack initiation in meta-implant designs[15] can serve as the foundation for patient-specific digital twins, while AI-enabled clinical decision systems extend this predictive capability into diagnosis and care planning. Realizing this vision will require close interdisciplinary collaboration among material scientists, biomedical engineers, data scientists, and clinicians.[15]

5.5 Research gaps and directions for clinical translation

Despite the breadth of demonstrated applications, several research gaps must be addressed before metamaterial-based healthcare technologies can move from laboratory prototypes to routine clinical use. First, fabrication remains a fundamental bottleneck across all metamaterial classes: the high-resolution lithography, laser sintering, and additive manufacturing routes needed to realize subwavelength architectures are costly, difficult to scale, and prone to imperfections that degrade device performance – for example, fabrication-induced broadening of the resonant response observed in terahertz silk-protein skin patches.[86] Future work should therefore prioritize reproducible, cost-effective, and standardized manufacturing protocols suitable for clinical-grade production.[16,19] Second, the long-term biocompatibility and in vivo safety of the metals, polymers, and nanostructures used in implantable and wearable metamaterials remain insufficiently characterized, since many promising systems – including degradable silk-based and acoustic-metamaterial patches – have so far been validated only in short-term, proof-of-concept animal studies;[86,88] systematic investigation of immune response, degradation behavior, and chronic tissue interaction is needed.[12,22] Third, although metamaterial biosensors and imaging systems achieve high sensitivity, their performance depends strongly on geometric and material parameters, so continued optimization of unit-cell design and exploration of alternative materials is required to ensure robust, reproducible specificity under real physiological conditions.[37,78] Fourth, clinical implementation will demand the integration of digital control, wireless communication, compact power sources, and closed-loop, sensor-triggered operation to enable real-time, personalized monitoring and therapy,[88] ideally coupled with AI-driven design and decision support.[100] Finally, the absence of standardized testing and regulatory frameworks for metamaterial-based devices continues to slow approval and commercialization; closing this gap will require close, sustained collaboration among materials scientists, clinicians, and regulatory bodies to translate these emerging platforms into validated, point-of-care tools.[20–23]

6. Clinical translation, clinical trials, and regularities considerations

While these technologies are early in the clinical translation process, there are significant hurdles covering manufacturing maturity, regulatory, and clinical validation. Scientific and engineering challenges to address in the regulation of AM-manufactured orthopedic implants include the source material, design, and postprinting characterization of the AM process, AM printing characteristics and characteristics of the printed product, physical and mechanical characterization of the final product, and cleaning, sterility and biocompatibility considerations.[16] The FDA has been collaborating with the commercial, clinical, and scientific community to offer guidance regarding AM in medical devices.[16] Although manufacturing fidelity is high in prototypes, the scalability and reproducibility of design innovations in a manufacturing environment continue to be an area of active research[15] and large-scale adoption of these computational and fabrication tools is still far off.[16]

There is also a lack of clinical evidence in comparison to preclinical promise. Indeed, efficacy has been well established in vitro and in animal models only in the nanomedicine field, and gold-nanoparticle-based theranostics for oral cancer have not yet entered the clinical trials or clinical applications.[35] An important issue is biosafety as conflicting in vivo toxicity data exists that needs to be resolved before clinical trials can be conducted.[35] These technical and trial readiness risks are compounded by the risks of long-term safety concerns, data privacy and security risks, potential biases in the algorithms involved in machine learning, and the need for regulation to be updated and refined to ensure compliance with patient safety requirements and ethical

guidelines.[15] Collectively, these are encouraging proof-of-concept results and while clinical applications in healthcare are still to be realized, there is a requirement for cross-disciplinary collaboration, as well as manufacturing, validation and regulatory development. Some researchers have also been suggested Ti-alloy materials[103–104] are used for medical additive products and frequency-based metamaterials[95–97] and functionally graded materials are used for bone products.

7. Conclusion

Metamaterials represent a paradigm shift in healthcare materials science, offering unparalleled control over physical interactions relevant to diagnosis, therapy, and biomedical devices. It is possible to achieve better imaging resolution, biosensing with extreme sensitivity, targeted cancer therapy, bio-machines, and responsive health sensors and implants through metamaterials-engineered architectures. The paper has given a comprehensive and organized knowledge on the role of metamaterials in healthcare, including the basics of the concept, the different types of materials, smart metamaterial systems, classical applications, general application areas, limitations, and opportunities. Although major issues associated with fabrication, biocompatibility, scalability, and regulation still exist, current interdisciplinary studies still push the limits of metamaterial innovation. As AI, additive manufacturing and bioinspiration design continue to evolve, metamaterials will be highly significant in the next generation of healthcare. Their effective clinical translation can greatly enhance the quality of diagnostics, treatment efficacy, and patient quality of life, which is the ambition of contemporary healthcare.

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